Distribution Fitting

From Discrete Quantile Forecasts to a Continuous Conditional Density

Methodological section continuing the direct forecasting framework of the previous chapter.

1 Motivation: Why a Single Quantile Is Not Enough

The direct forecasting model delivers, for every forecast origin t and horizon h, a finite set of predicted conditional quantiles:

$$\hat{Q}_{t+h} = \left[\hat{q}_{t+h}(\tau_1),\ \hat{q}_{t+h}(\tau_2),\ \ldots,\ \hat{q}_{t+h}(\tau_K)\right], \qquad \tau_k \in \{0.05,\, 0.25,\, 0.50,\, 0.75,\, 0.95\}. \tag{1}$$

These point forecasts are the discrete skeleton of the conditional distribution of future Brent returns. A portfolio manager who needs only a single Value-at-Risk threshold at the 5th percentile can read $\hat{q}_{t+h}(0.05)$ directly and stop. But the ambitions of the Oil-at-Risk framework extend well beyond a single threshold.

Consider what a central bank or a sovereign wealth fund actually needs when a geopolitical crisis unfolds in the Persian Gulf. The question is not merely whether the 5th percentile of the return distribution has moved; the question is how the entire shape of the conditional density has changed. Has the left tail fattened while the right tail remained stable, signalling a pure downside vulnerability? Has the distribution become bimodal, reflecting the market's inability to resolve whether a supply disruption will materialise? Has skewness reversed, as it does when a demand-side collapse overtakes a supply-side fear? To compute Expected Shortfall, one must integrate below the quantile. To measure Kullback–Leibler divergence between successive conditional densities, one must have a density. To study how geopolitical shocks reshape the skewness and kurtosis of the forecast distribution across horizons, one must recover the full shape of the curve, not merely a handful of points on it. The discrete quantile skeleton must therefore be interpolated into a continuous parametric density that preserves the information encoded in the quantile forecasts while supplying the analytic tractability that downstream tail-risk diagnostics require.

2 The Unconditional Approach and Its Limitations

A natural first instinct is to fit a distribution to the historical return series by Maximum Likelihood and use it as the forecast distribution. This is, in fact, the standard procedure in traditional risk management. The analyst collects the full sample of observed Brent log-returns, orders them, computes their empirical density, and maximises the log-likelihood over the parameter space of the chosen family. The resulting density captures the long-run statistical properties of the return-generating process: its average volatility, its historical skewness, and its unconditional tail weight.

This unconditional fit is perfectly legitimate when the objective is descriptive: characterising what the distribution of oil returns has looked like over the past three decades. In commodity risk management practice, the Value-at-Risk over an n-day horizon is routinely obtained by fitting such a density to historical returns and scaling the resulting quantile by $\sqrt{n}$, or equivalently by computing returns over non-overlapping n-day windows and fitting the density to that lower-frequency series. The procedure is tractable, well understood, and backed by large sample sizes.

Yet it suffers from a fundamental limitation: it is entirely backward-looking. The density estimated by Maximum Likelihood over the full history is an average over all regimes the market has traversed, from the relative calm of the mid-2010s to the extreme dislocations of the 2020 pandemic collapse and the 2022 invasion of Ukraine. It cannot tell the risk manager whether today's geopolitical conditions make the left tail fatter or thinner than usual. A Strait-of-Hormuz escalation and a period of de-escalation produce the same unconditional density, because the conditioning information is averaged away.

The Oil-at-Risk framework is explicitly conditional. The quantile forecasts in (1) already encode the current state of the world through the regressors: the seven-day moving average of geopolitical risk, the lagged realised volatility, the dollar index, and the Baltic Dry Index. The distributional fitting step must respect this conditioning rather than overwrite it with a time-invariant density. What we need is not a single distribution estimated once over the full history, but a sequence of distributions, one for each forecast origin, each shaped by the covariates prevailing at that origin.

3 The Johnson S_U Distribution

The choice of parametric family must satisfy two requirements. First, it must be flexible enough to accommodate the stylised facts of crude oil returns: excess kurtosis, asymmetric tails, and time-varying skewness whose direction shifts as the geopolitical regime evolves. When a supply disruption threatens, the right tail of the return distribution fattens as the market prices in potential price spikes; when a demand collapse materialises, the left tail fattens instead. A distribution that imposes symmetry, or that controls tail weight through a single parameter, cannot track these asymmetric rotations. Second, its cumulative distribution function must admit a closed-form inverse, because the estimation procedure described in Section 4 evaluates the theoretical quantile function $F^{-1}(\tau \mid \Theta)$ at every iteration of the numerical optimiser, and a distribution requiring numerical root-finding at each evaluation would render the fitting computationally prohibitive over thousands of rolling-window origins.

The Johnson S_U distribution, introduced by Johnson (1949) as part of a system of frequency curves generated by translation of the standard normal, meets both criteria. The core idea is elegant: if Z is a standard normal random variable, the transformation

$$Z = \gamma + \delta \sinh^{-1}\left(\frac{X-\xi}{\lambda}\right), \qquad Z \sim \mathcal{N}(0,1), \tag{2}$$

maps X into normality through the inverse hyperbolic sine. The parameter vector $\Theta = \{\gamma, \delta, \xi, \lambda\}$ governs, respectively, the skewness, the tail weight, the location, and the scale of the distribution. By inverting the transformation, X inherits from the normal a full analytic apparatus while being free to exhibit heavy tails and pronounced asymmetry. The probability density function of X is

$$f(x \mid \Theta) = \frac{\delta}{\lambda\sqrt{2\pi}} \frac{1}{\sqrt{1+\left(\frac{x-\xi}{\lambda}\right)^2}} \exp\left\{-\frac{1}{2}\left[\gamma + \delta \sinh^{-1}\left(\frac{x-\xi}{\lambda}\right)\right]^2\right\}, \tag{3}$$

and its cumulative distribution function reduces to

$$F(x \mid \Theta) = \Phi\left(\gamma + \delta \sinh^{-1}\left(\frac{x-\xi}{\lambda}\right)\right), \tag{4}$$

where $\Phi(\cdot)$ is the standard normal CDF. The quantile function is obtained by direct inversion of (4):

$$F^{-1}(\tau \mid \Theta) = \xi + \lambda \sinh\left(\frac{\Phi^{-1}(\tau)-\gamma}{\delta}\right). \tag{5}$$

This closed-form quantile function is the decisive computational advantage: it evaluates in microseconds, involves no numerical root-finding, and makes the Minimum Distance Estimator described below as fast as a simple least-squares problem.

The economic appeal of the S_U family becomes clear when one considers the alternative. The Generalised Error Distribution, widely used in GARCH modelling, controls tail weight through a single shape parameter but imposes symmetry: it cannot distinguish between a left-skewed distribution during a demand collapse and a right-skewed one during a supply panic. The Johnson S_U provides independent control over skewness and kurtosis through two separate shape parameters, γ and δ. In the limit $\gamma \to 0$, $\delta \to \infty$, the S_U nests the Gaussian as a special case, so that tranquil periods are accommodated without forcing heavy tails where none exist. The original Growth-at-Risk framework of Adrian, Boyarchenko, and Giannone (2019) employs the Azzalini skewed-t distribution for this step; the S_U represents a more finance-oriented alternative that has been used successfully to model asset returns in portfolio management and Value-at-Risk contexts, with comparable flexibility and a simpler quantile-function inversion.

4 Minimum Distance Estimation

In the classical Maximum Likelihood setting described in Section 2, the analyst observes thousands of realisations from the return-generating process, each one an independent draw from the stochastic process whose density is to be estimated. The likelihood surface is well identified, the sample is large, and the estimator is efficient.

The setting that arises after the quantile regression step is fundamentally different in two respects. First, for a given forecast origin t and horizon h, we do not observe a large sample of future returns. We observe a small vector of K predicted quantiles, one for each targeted quantile level. With $K = 5$, we have five data points, not five thousand. Second, and more critically, these five points are not realisations of the random variable whose distribution we seek to characterise. They are *quantiles* of that distribution, estimated with noise from the regression. A likelihood function cannot be written over quantiles in the standard sense, because quantiles are functionals of the distribution, not data points drawn from it. Attempting to treat the predicted quantiles as if they were observed returns and maximising a likelihood would produce a density that is internally inconsistent with the quantile regression that generated the inputs.

Minimum Distance Estimation resolves this mismatch by reformulating the fitting problem as a function-matching exercise. The question it answers is direct: find the parameter vector $\hat{\Theta}_t$ such that the theoretical quantile function of the Johnson S_U distribution reproduces the empirical quantile forecasts as closely as possible. Formally, the estimator solves

$$\hat{\Theta}_t = \arg\min_{\Theta} \sum_{\tau \in \mathcal{T}} \Big(\hat{q}_{t+h}(\tau) - F^{-1}(\tau \mid \Theta) \Big)^2, \tag{6}$$

where $\mathcal{T} = \{0.05, 0.25, 0.50, 0.75, 0.95\}$ is the set of targeted quantile levels and $F^{-1}(\cdot \mid \Theta)$ is the Johnson S_U quantile function given by (5). The objective is the Sum of Squared Errors between the forecasted and theoretical quantiles: the algorithm searches for the S_U curve whose quantiles at $\tau = 0.05, 0.25, 0.50, 0.75$, and 0.95 lie as close as possible to the five quantile forecasts produced by the regression.

Because (6) has no closed-form solution, it must be solved numerically. The optimiser iteratively proposes candidate values for $\Theta = \{\gamma, \delta, \xi, \lambda\}$, evaluates the S_U quantile function at each $\tau \in \mathcal{T}$ using the closed-form expression (5), computes the SSE, and adjusts the parameters until convergence. We employ the L-BFGS-B algorithm (Byrd, Lu, Nocedal, and Zhu, 1995), a limited-memory quasi-Newton method that supports box constraints on the parameters. Bounding the search space is essential: δ and λ must remain strictly positive to preserve a valid density, and unconstrained optimisation risks divergence in the tails of the parameter space, particularly during crisis episodes when the predicted quantiles are spread far apart and the loss surface becomes flat over wide regions.

Two algorithmic choices materially affect the reliability of the solution. First, the initial guess Θ_0 strongly influences the speed and likelihood of convergence. We initialise ξ at the predicted median $\hat{q}_{t+h}(0.50)$, λ at the interquartile range $\hat{q}_{t+h}(0.75) - \hat{q}_{t+h}(0.25)$, and γ and δ at zero and one, respectively. This encodes a symmetric, moderately heavy-tailed prior centred on the regression's median forecast, a sensible starting point that the optimiser then refines as it absorbs the asymmetry embedded in the outer quantiles. Second, convergence tolerances must be tight enough to produce an accurate fit but loose enough to avoid excessive iterations on nearly flat regions of the SSE surface; a function tolerance of 10^{-10} has proven adequate across the full out-of-sample experiment.

The procedure is repeated independently for every forecast origin in the rolling window, yielding a time series of fitted parameter vectors $\{\hat{\Theta}_t\}_{t \in \mathcal{W}}$ and, through them, a time series of continuous conditional densities. Each density is anchored to the information set available at its forecast origin: it inherits the conditioning embedded in the quantile forecasts and imposes no cross-sectional pooling across dates. The result is a sequence of distributions that evolves with the geopolitical and macroeconomic environment, fattening its left tail when the Geopolitical Risk index spikes and narrowing it when conditions normalise, precisely the object the Oil-at-Risk framework requires.

5 Evaluation: The Probability Integral Transform

Once the Minimum Distance Estimator has produced a fitted conditional distribution for every forecast origin, the critical question becomes whether those distributions are accurate descriptions of the future reality they purport to model. A distribution that fits its own quantile inputs perfectly but systematically misrepresents the realised returns is operationally worthless: a risk manager who trusts its Expected Shortfall is taking on precisely the exposure she believes she has hedged. The evaluation framework must therefore confront the fitted densities with out-of-sample data that were never used to estimate either the quantile regressions or the MDE parameters.

The foundational diagnostic is the Probability Integral Transform (PIT), formalised in the context of density forecast evaluation by Diebold, Gunther, and Tay (1998). For each forecast origin t in the out-of-sample window, the PIT evaluates the realised return y_{t+h} under the conditional CDF that was forecasted for that date:

$$u_t = F(y_{t+h} \mid \hat{\Theta}_t). \tag{7}$$

The transformed value u_t is the cumulative probability of observing a return equal to or less than the realised value y_{t+h} according to the forecasted conditional distribution. Its interpretation maps directly onto the economic events that the Oil-at-Risk framework is designed to capture. If y_{t+h} was a large unexpected decline, a geopolitical supply disruption or a demand-side rout that the model did not fully anticipate, it falls deep in the left tail of the forecasted distribution and u_t takes a value close to zero. If y_{t+h} was a massive bull run, perhaps driven by a de-escalation that relaxed risk premia faster than the model expected, it falls in the right tail and u_t approaches one. If the model anticipated the realised value accurately, assigning it a probability mass near the centre of the distribution, u_t falls near the middle of the unit interval.

5.1 Theoretical justification

The theoretical result that underwrites the PIT diagnostic is sharp and requires only the assumption that the forecasted CDF is continuous. Suppose the forecasted CDF F is identical to the true data-generating CDF. Then the random variable $U = F(Y)$ is uniformly distributed on $[0, 1]$. The proof proceeds in three lines. Write the CDF of U:

$$P(U \leq u) = P(F(Y) \leq u). \tag{8}$$

Because F is a continuous, strictly non-decreasing function, its inverse F^{-1} exists and preserves the direction of the inequality:

$$P\big(F(Y) \leq u\big) = P\big(Y \leq F^{-1}(u)\big). \tag{9}$$

By the definition of the cumulative distribution function, $P\big(Y \leq F^{-1}(u)\big) = F\big(F^{-1}(u)\big) = u$, so

$$P(U \leq u) = u \quad \text{for all } u \in [0, 1], \tag{10}$$

which is the defining property of the Uniform$(0, 1)$ distribution.

The contrapositive supplies the diagnostic: if the sequence of PITs $\{u_t\}$ departs from uniformity, the forecasted distribution is misspecified. Departures manifest in characteristic ways that carry direct economic meaning. An excess of PITs near zero indicates that the model's left tail is too thin: it underestimates the probability of large negative returns, precisely the kind of miscalibration that would lead a risk manager to understate her Expected Shortfall during a geopolitical crisis. An excess near one indicates a right tail that is too thin, underestimating the probability of large price spikes. A hump in the centre of the PIT histogram signals that the distribution is too dispersed, assigning too much probability to extreme outcomes that rarely materialise, a conservative miscalibration that is costly in a different way because it overstates risk and forces unnecessarily aggressive hedging. A U-shaped histogram signals the opposite: the distribution is too narrow, and both tails are systematically surprised by realisations the model deems improbable.

5.2 The Kolmogorov–Smirnov test on the PITs

The PIT histogram provides a visual impression of calibration quality; the Kolmogorov–Smirnov (KS) test furnishes a formal statistical assessment. The KS statistic measures the supremum absolute deviation between the empirical CDF of the PIT sequence and the theoretical CDF of the Uniform$(0, 1)$ distribution:

$$D_n = \sup_{u \in [0,1]} \big|\hat{F}_n(u) - u\big|, \tag{11}$$

where $\hat{F}_n(\cdot)$ is the empirical CDF of the n out-of-sample PITs. Under the null hypothesis that the PITs are drawn from a Uniform$(0, 1)$ population, the distribution of D_n is distribution-free and tabulated for all sample sizes. Rejection of the null at the chosen significance level constitutes evidence that the fitted conditional distributions are systematically miscalibrated: the model's probabilistic characterisation of future oil returns does not match what actually happens.

The distinction between in-sample and out-of-sample evaluation is critical here, and it is worth pausing to explain why. If the PITs are computed from the same data used to estimate the quantile regression and the MDE parameters, the fitted distributions are mechanically close to the data by construction. The quantile regression has been optimised to place the quantile boundaries in the right locations relative to the training returns, and the MDE has been optimised to thread a smooth curve through those boundaries. Testing whether the resulting PITs are uniform is then akin to asking whether a model fits its own training data, a question whose answer is almost always yes and is therefore uninformative. This is an instance of the Lilliefors effect: testing goodness-of-fit against a distribution whose parameters were estimated from the same sample systematically deflates the true size of the test. In a Monte Carlo experiment, the in-sample KS test would reject far less often than its nominal α level suggests, because the estimation has already consumed the degrees of freedom that the test needs to detect departures.

For this reason, we compute the PIT sequence exclusively over the out-of-sample window, where the realised returns y_{t+h} were never used in the estimation of either the quantile regression coefficients or the MDE parameters. In this setting, the standard KS critical values apply without adjustment, and the test possesses its nominal size.

5.3 Design of the evaluation

The PIT evaluation is conducted over the same out-of-sample window used for the coverage backtests in the direct forecasting chapter, beginning in January 2023 and extending to April 2026. For each trading day t in this window, the rolling-window quantile regressions produce a vector of predicted quantiles $\hat{Q}_{t+h}$ at the target horizon h; the MDE fits a Johnson S_U distribution to those quantiles, yielding $\hat{\Theta}_t$; and the PIT $u_t = F(y_{t+h} \mid \hat{\Theta}_t)$ is recorded. The resulting sequence of PITs is then examined both visually, through a histogram binned over $[0, 1]$, and formally, through the KS test of (11).

This design constitutes a single, continuous goodness-of-fit test that spans the entire out-of-sample period. It does not evaluate the distribution at a single date or at a single quantile level; it evaluates every fitted density across every forecast origin simultaneously, asking whether the sequence of conditional distributions, taken as a whole, is a statistically adequate model of the data-generating process observed over more than three years of out-of-sample Brent returns. The test is therefore considerably more demanding than a pointwise coverage backtest, which asks only whether the correct fraction of returns fall below a given quantile. The PIT test asks whether the entire shape of the conditional density, its location, its spread, its skewness, and its tail weight, is correctly specified at every date.

From an economic standpoint, this is precisely the validation that the Oil-at-Risk framework requires. The downstream risk measures that give the framework its operational value, Expected Shortfall, Kullback–Leibler divergence, and the entropy-based early warning system, all depend on the shape of the full conditional density, not merely on a single quantile threshold. If the PIT test rejects, those measures inherit the misspecification and their economic interpretation becomes unreliable: an Expected Shortfall computed from a miscalibrated density is not the true expected loss in the tail, and a KL divergence measured against a misspecified baseline does not capture the genuine informational shift in the conditional distribution. Passing the PIT test is therefore the necessary condition for treating the fitted distributions as a credible probabilistic characterisation of future oil price vulnerabilities under geopolitical duress.

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